MIL-P-13822 (Ord)
6 December 1954
SUPERSEDING
MA-PD-526
7 May 1954

MILITARY SPECIFICATION

POLYISOBUTYLENE/OIL GEL

1. SCOPE

1.1 This specification covers polyisobutylene/oil gel intended for use as a component of Military dynamite.

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

MILITARY

MIL-L-2104 - Lubricating Oil, Internal-Combustion Engine, Heavy Duty.
MIL-C-2550 - General Specification For Ammunition Except Small Arms Ammunition.
MIL-P-13298 - Polyisobutylene

STANDARDS

MILITARY

MIL-STD-129 - Marking For Shipment and Storage.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

2.2 Other publications.- The following document forms a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

INTERSTATE COMMERCE COMMISSION
49 CFR 71-78 - Interstate Commerce Commission Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles.

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(The Interstate Commerce Commission regulations are now a part of the Code of Federal Regulations (1949 Edition-Revised 1950), available from the Superintendent of Documents, Government Printing Office, Washington 25, D.C. Orders for the above publication should cite "49 CFR 71-78 (Rev. 1950)".)

3. REQUIREMENTS

3.1 Material.-The polyisobutylene/oil gel shall be a thoroughly homogeneous mixture of polyisobutylene, complying with Specification MIL-P-13298, incorporated with Grade 10 engine oil, complying with Specification MIL-L-2104.

3.2 Composition.- The polyisobutylene/oil gel, shall comply with the following composition, when determined as specified in 4.4.1 and 4.4.2:

Polyisobutylene, percent	7.5 ± 0.5
Grade 10 engine oil, percent	92.5 ± 0.5

3.3 Inorganic insoluble matter.-The inorganic insoluble matter shall be 0.05 percent, maximum when determined as specified in 4.5.

3.4 Acidity.-The acidity, as sulfuric acid, shall be 0.05 percent, maximum, when determined as specified in 4.6.

3.5 Workmanship.-The polyisobutylene/oil gel shall be free from foreign matter and visible impurities.

4. QUALITY ASSURANCE PROVISIONS

4.1 Lot.-A lot shall consist of material from the same batch or blending operation, subjected to the same processing operations and conditions.

4.2 Sampling.-An 8-ounce sample shall be selected from each lot and placed in an airtight container, and labeled to show the name of the material, manufacturer, plant, contract or purchase order number, lot number, and lot size. All acceptance tests shall be made on this sample.

4.3 Inspection.-Inspection shall be as specified in Specification MIL-S-2550.

4.4 Determination of composition.-

4.4.1 Polyisobutylene content.-An accurately weighed portion of approximately 2 gm of the sample material shall be transferred to a 150-ml beaker and 30 ml of 1:1 ether-acetone solution added. The beaker and contents shall be placed on a steam bath and allowed to heat gently while covered with a watch glass. After 2 minutes the beaker and contents shall be removed from the steam bath. With the aid of a glass stirring rod, flattened at one end, the sample shall be thoroughly worked to assure complete penetration of the solvent into the mass.

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After allowing the solution to cool, the solution shall be decanted into a tared medium-porosity sintered glass crucible. This procedure shall be repeated twice making a total of three treatments. An additional 30 ml of ether-acetone solution shall be added, and the remaining polyisobutylene transferred to the crucible with the aid of the stirring rod. The beaker shall be thoroughly rinsed with a stream of ether-acetone solution, allowing the rinsings to run into the crucible. Suction shall be applied to the crucible until the odor of ether is no longer detectable. The crucible and contents shall be dried in an oven maintained at 100° ± 5°C for 1 hour. The crucible and contents shall be cooled in a desiccator and weighed. The polyisobutylene content shall be calculated as follows:

$$\text{Polyisobutylene, percent} = \frac{(A-B) \times 100}{W}$$

where:

A = Weight of the crucible and residue.
B = Weight of the empty crucible.
W = Weight of the sample.

4.4.2 Oil content.- The percent oil content shall be determined by difference; i.e. the percentage of polyisobutylene and inorganic insoluble matter (determined as specified in 4.5) shall be added, and the sum subtracted from 100 percent.

4.5 Determination of inorganic insoluble matter.-An accurately weighed portion of approximately 10 gm of the sample shall be transferred to a 600-ml beaker and 300 ml of benzene shall be added. The beaker and contents shall be placed on a steam bath. The sample shall be heated until it is completely dissolved. The benzene solution shall be filtered through a tared medium porosity porcelain crucible, which has been previously ignited to 600°C. Care must be taken to insure complete transfer of the undissolved portion to the crucible. The crucible and contents shall be washed with 30 ml of benzene. The crucible shall be aspirated until the odor of benzene is no longer detectable. The crucible shall then be placed in a muffle furnace at 600°C for 1 hour. The crucible shall be removed from the furnace and cooled to room temperature and weighed. The percentage of inorganic insoluble matter shall be calculated as follows:

$$\text{Inorganic insoluble matter, percent} = \frac{(A-B)\ 100}{W}$$

where:

A = Weight of crucible and residue.
B = Weight of empty crucible.
W = Weight of the sample.

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4.6 Determination of acidity (as sulfuric acid).-An accurately weighed portion of approximately 5 gm of the sample shall be transferred to a 400-ml beaker. Seventy-five milliliters of toluene shall be added and the beaker and contents heated on a steam bath, with occasional stirring, until the sample appears to be completely dissolved. Seventy-five milliliters of alcohol, which has been previously neutralized with 0.1N sodium hydroxide using 0.5 ml of phenolphthalein as the indicator, shall be added to the solution with constant stirring, then 75 ml of distilled water shall be added. The resulting mixture shall be titrated, using 0.1N sodium hydroxide solution, to a light pink end point. The percentage of acidity shall be calculated as follows:

$$\text{Percentage of acidity (as sulfuric acid)} = \frac{4.9\ AN}{W}$$

where:

A = ml of sodium hydroxide used in the titration.
N = Normality of the sodium hydroxide solution.
W = Weight of sample.

4.7 Resubmission and retest.-If the sample subjected to test, fails to pass the test, the lot shall be rejected subject to reworking and resubmission. The lot shall be considered acceptable provided that a new sample selected in accordance with 4.2 passes all the tests required by this specification.

5. PREPARATION FOR DELIVERY

5.1 Packing.-Polyisobutylene/oil gel shall be packed in removable head steel containers conforming to Interstate Commerce Commission Specifications 6A or 6B stipulated in the Code of Federal Regulations 49 CFR 71-78.

5.2 Marking.-In addition to any special marking required by the contract or purchase order, shipments shall be marked in accordance with Standard MIL-STD-129.

6. NOTES

6.1 Ordering data.-Procurement documents should specify the title, number, and date of this specification.

Notice.-When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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